Organic Chemistry Part 2

CHAPTER 1

Grignard Reagents and Organometallic Compounds **1.1–1.4**

DPP 1.1 Grignard Reagents and Organometallic Compounds 1.1

Ananth Garg

CHAPTER 1

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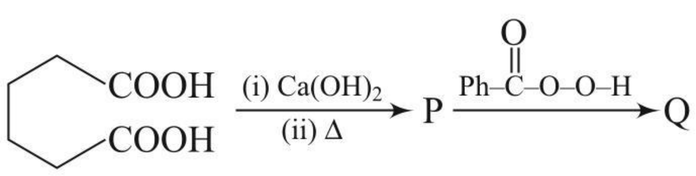
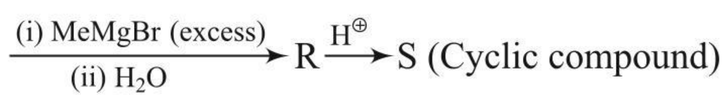
DPP 1.1

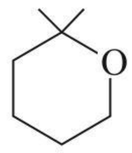
Grignard Reagents and Organometallic Compounds

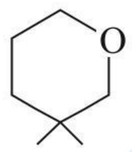
Single Correct Answer Type

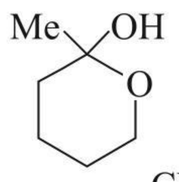
1. 

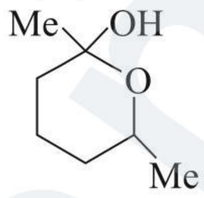


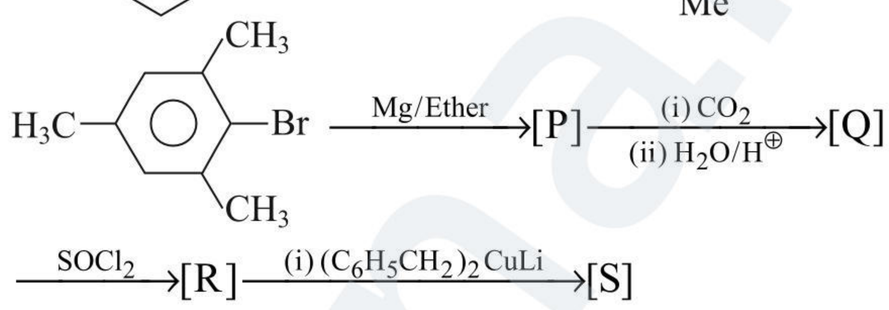
The cyclic compound S is

(a) 

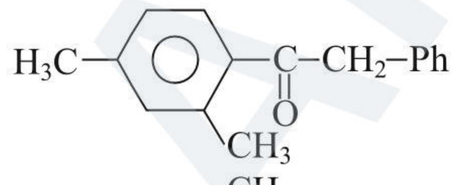
(b) 

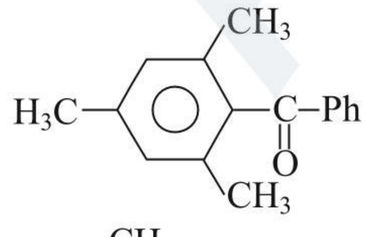
(c) 

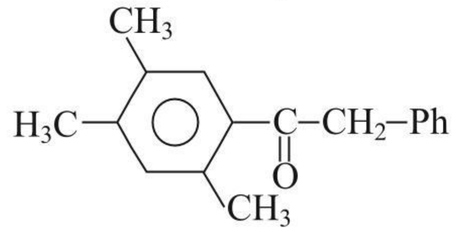
(d) 

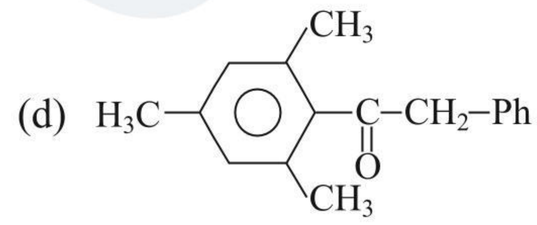
2. 

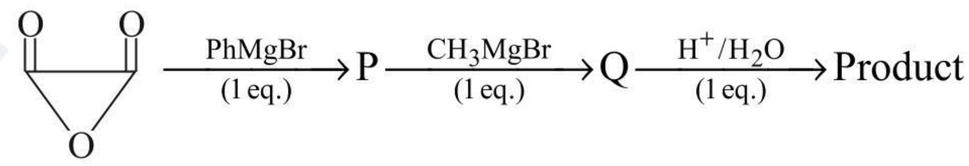
Identify (S) in the following sequence of reaction.

(a) 

(b) 

(c) 

(d) 

3. 

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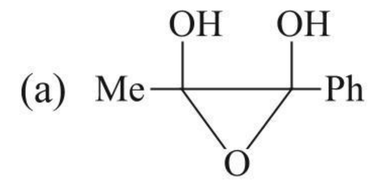
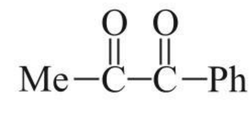
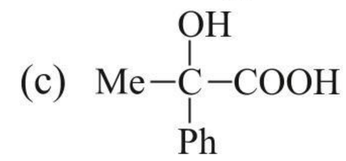
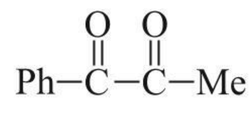
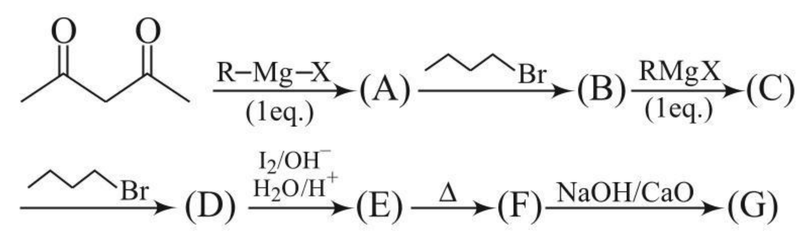
(a) 

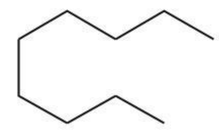
(b) 

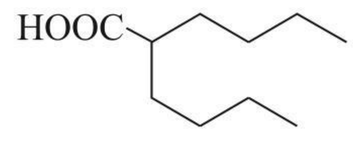
(c) 

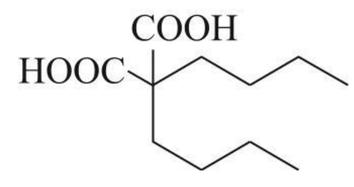
(d) 

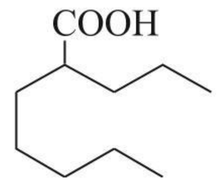
4. 

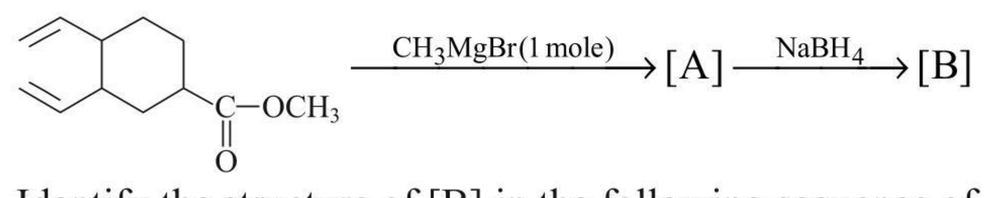
Identify the structure of 'G'.

(a) 

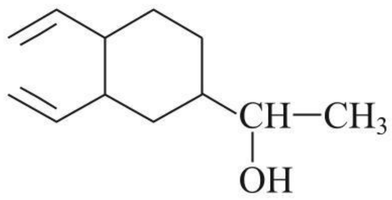
(b) 

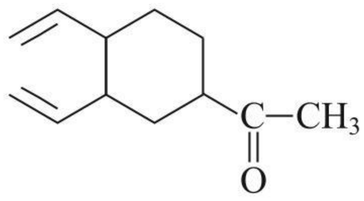
(c) 

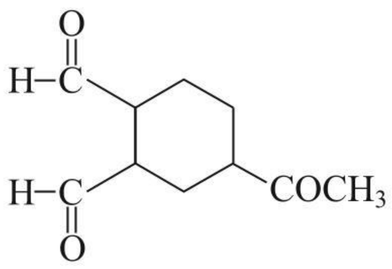
(d) 

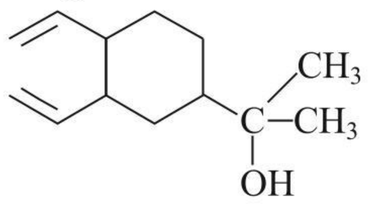
5. 

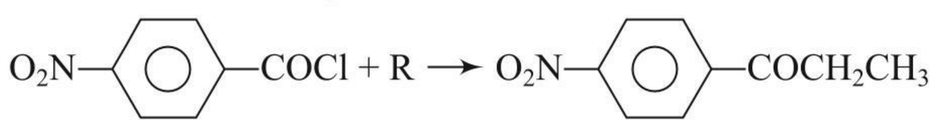
Identify the structure of [B] in the following sequence of reactions.

(a) 

(b) 

(c) 

(d) 

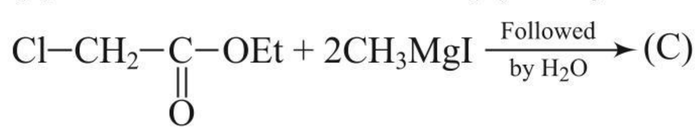
6. 

The reagent R may be

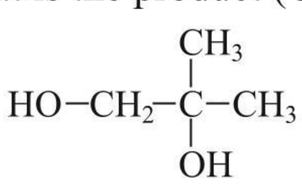
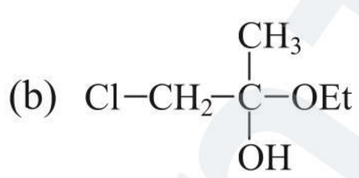
CH_3CH_2MgBr (I) $\quad$ CH_3CH_2Li (II)

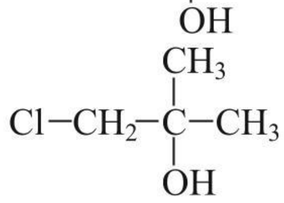
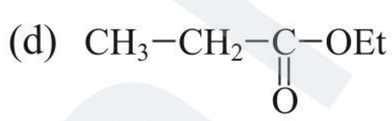
$(CH_3CH_2)_2Cd$ (III) $\quad$ $(CH_3CH_2)_2CuLi$ (IV)

(a) I or II (b) I or II or III
(c) III or IV (d) Any of the four

7. 

What is the product (C)?

(a)  (b) 

(c)  (d) 

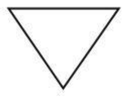
8. $2CH_2 = CH - CH_2 - Cl \xrightarrow[\text{Ether}]{Mg} A \xrightarrow[\text{Followed by } H_2O]{B}$

$(CH_2 = CH - CH_2)_2CH - OH + CH_3 - CH_2 - OH$

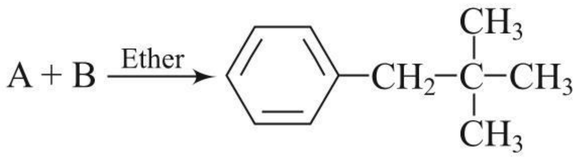
In the above chemical reaction, B may be

(a) Methyl formate (b) Ethyl formate
(c) Isopropyl methanoate (d) Ethanoyl chloride

9. The reaction, $BrCH_2CH_2CH_2Br + Mg \xrightarrow{Et_2O}$ produces mainly

(a) $CH_3CH_2CH_3$ (b) $BrMgCH_2CH_2CH_2MgBr$
(c) $CH_3CH = CH_2$ (d) 

10. The best yield of given product can be obtained by using which set of reactants A and B, respectively,



(a) PhLi + Neopentyl chloride
(b) *t*-Bu-MgBr + Benzyl bromide
(c) PhMgBr + Neopentyl bromide
(d) Benzylchloride + *t*-Butyl chloride

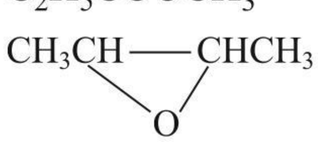
11. Which of the following is an organometallic compound?

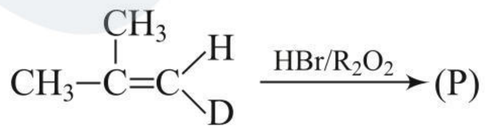
(a) Lithium methoxide
(b) Lithium acetate
(c) Lithium dimethylamide
(d) Methyllithium

12. Grignard reagent is not prepared in aqueous medium but prepared in ether medium, because

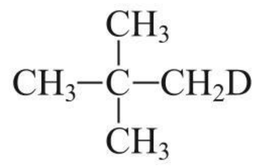
(a) the reagent is highly reactive in ether
(b) the reagent becomes inactive in water
(c) the reagent reacts with water
(d) none of three is correct.

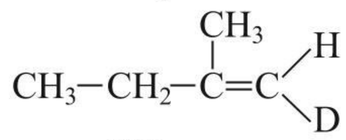
13. Which of the following compounds on reaction with CH_3MgBr will give a tertiary alcohol?

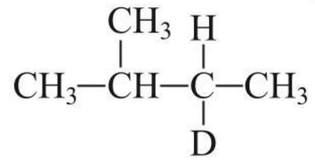
(a) C_6H_5CHO (b) $C_2H_5COOCH_3$
(c) C_2H_5COOH (d) 

14. (i) 

(ii) P (dextrorotatory) $\xrightarrow{CH_3MgBr}$ (Q). The end product (Q) is

(a) 

(b)  (single stereoisomer)

(c)  (single stereoisomer)

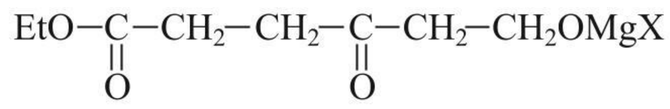
(d) A racemic mixture of $(CH_3)_2CH–CHD–CH_3$

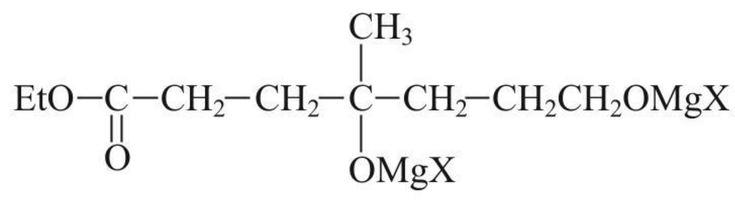
Multiple Correct Answers Type

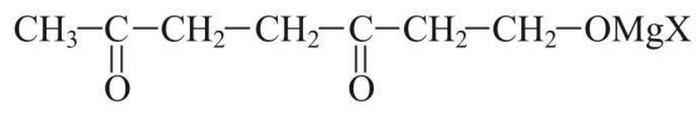
15. Analyse the following reaction

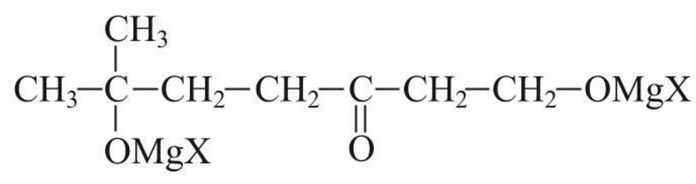
$HOCH_2CH_2COCH_2CH_2CO_2Et$

$\xrightarrow{1CH_3MgX} A \xrightarrow{1CH_3MgX} B \xrightarrow{1CH_3MgX} C$ if

(a) 

(b) 

(c) 

(d) 

Which is/are correct?

(a) A is a (b) B is c
(c) C is not d (d) B is b

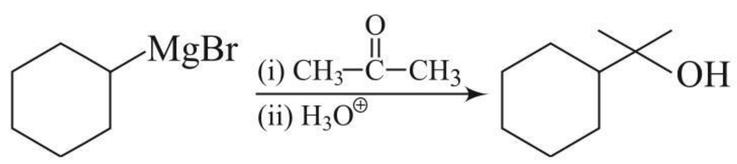
16. Which of the following compounds formed by the reaction of *sec*-butylmagnesium iodide with a terminal alkyne?

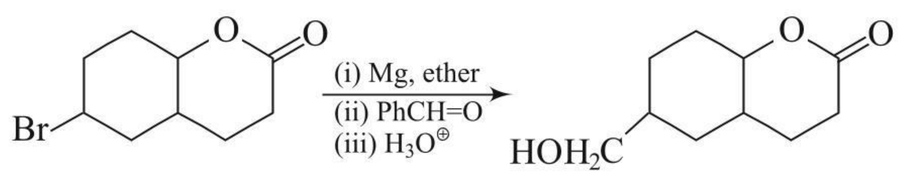
(a) *n*-Butane (b) Isobutane
(c) Higher alkyne (d) Unsaturated Grignard's reagent

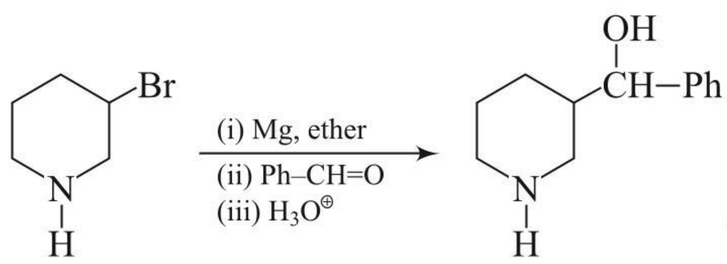
17. *n*-Propyl alcohol is obtained on hydrolysis of the adduct obtained by the reaction of

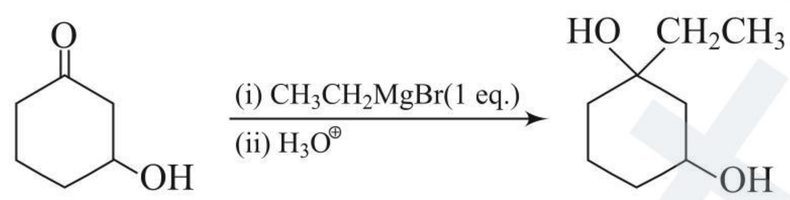
(a) EtMgX and HCHO (b) MeMgX and CH_3CHO
(c) EtMgX and O_2 (d) MeMgX and $(CH_2)_2O$

18. Which of the following is/are the incorrect Grignard reaction?

(a) 

(b) 

(c) 

(d) 

19. Which of the following statement(s) is (are) true?

(a) Sodium forms ionic organometallic compounds.
(b) Sodium does not form any organometallic compound.
(c) Mercury forms ionic organomercury compounds
(d) Mercury forms covalent organomercury compounds.

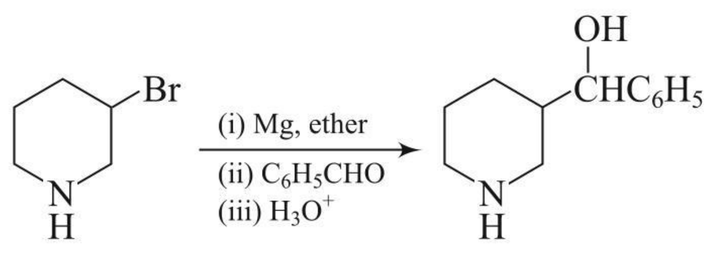
20. Solvent used during preparation of organolithium compounds should be completely dry because

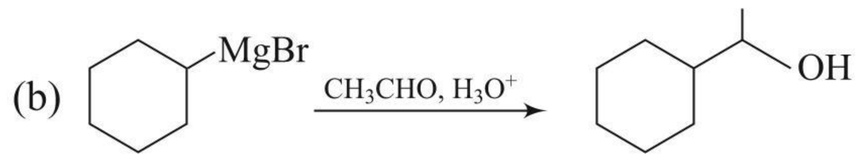
(a) lithium forms insoluble hydroxide
(b) the organolithium compound formed will react with water
(c) the solvent is protonated and thus becomes inactive
(d) solvent is insoluble in water.

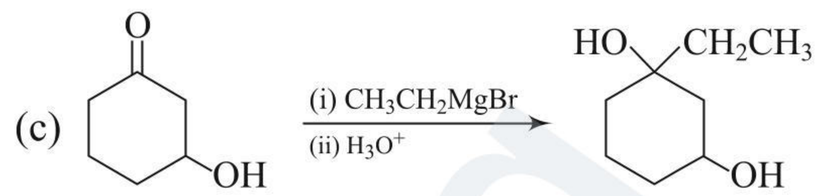
21. Which of the following statement(s) is/are true?

(a) Methylmagnesium iodide is a powerful Bronsted base.
(b) Methylmagnesium iodide is a powerful Lewis base.
(c) Methylmagnesium bromide is a powerful nucleophile.
(d) Methylmagnesium bromide is a weak nucleophile.

22. Which of the following reaction(s) is (are) not feasible?

(a) 

(b) 

(c) 

(d) All the three are feasible

Matching Column Type

23. Match the items given in column I with those in column II.

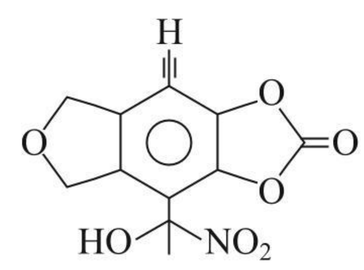
	Column I		Column II
a.	RMgI + Acetonitrile ($CH_3C\equiv N$)	p.	Alkanone
b.	RMgI + Carbon disulphide	q.	Ester
c.	RMgI . (1 eq) + Ethyl chloroformate	r.	1° Alcohol
d.	RMgI + Oxirane	s.	Dithionic acid

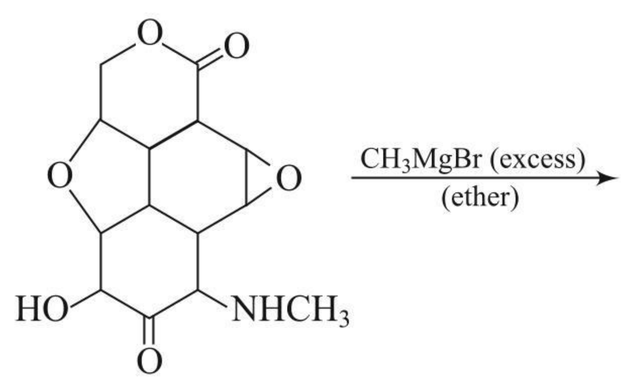
24. Match the items given in column I with those in column II.

	Column I		Column II
a.	CH_3MgBr + epoxide	p.	Zerewitinoff estimation
b.	$C_6H_{11}MgCl + CH_2O$	q.	2° alcohols
c.	Solvent in preparing RMgX	r.	Pentane
d.	$n\text{-}C_5H_{11}MgBr + NH_3$	s.	1° alcohols

Numerical Value Type

25. Calculate the number of molecules of Grignard reagent consumed by 1 molecule of following compound.



26. 

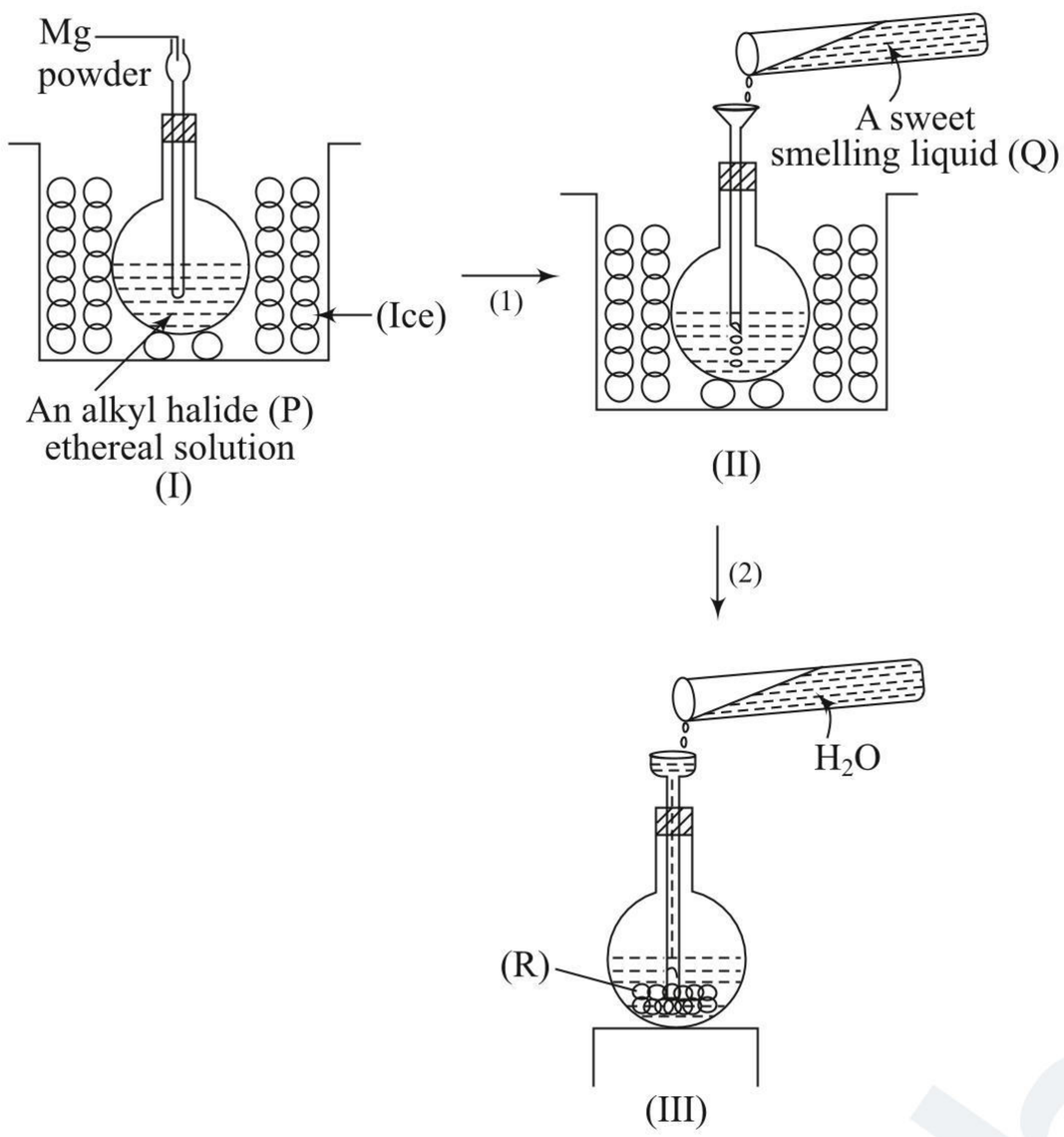
Find the number of CH_3MgBr molecules used by the given reactant, to react completely.

27. How many total possible Grignard reagent can give 3-methylpentane on treatment with $H^{\oplus}$?

Linked Comprehension Type

For questions 28-30

Observe the following experiment



28. If the reactant 'P' is ethyl chloride, then the product R can be

(a) $CH_3-C(CH_3)(OH)-CH_3$ (b) $CH_3CH_2-O-CH_2CH_3$

(c) $CH_3-C(CH_2CH_3)(OH)-CH_2-CH_3$ (d) $CH_3CH_2-C(CH_3)(OH)-CH_3$

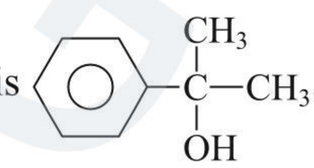
29. If the liquid Q is $H-\underset{\underset{O}{\|}}{C}-OC_2H_5$, then the product R can be (P can be any other halide)

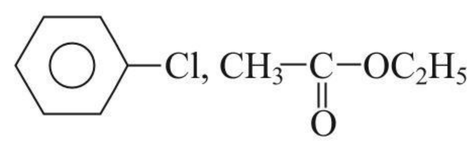
(a) $CH_3-CH(OH)-CH_3$

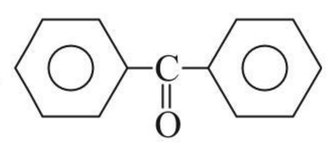
(b) $CH_3-C(CH_3)(OH)-CH_3$

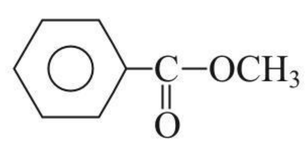
(c) $H-C(H)(OH)-C_2H_5$

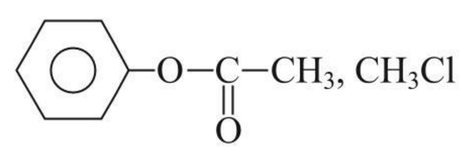
(d) $C_2H_5-C(C_2H_5)(OH)-C_2H_5$

30. If R is  , then P and Q can be respectively

(a) 

(b) $CH_3-CH(Cl)-CH_3$, 

(c) CH_3-Cl, 

(d) 

ANSWER KEY

Single Correct Answer Type

1. (a)	**2.** (d)	**3.** (c)	**4.** (a)	**5.** (a)
6. (c)	**7.** (c)	**8.** (b)	**9.** (d)	**10.** (b)
11. (d)	**12.** (c)	**13.** (b)	**14.** (c)	

Multiple Correct Answers Type

15. (a, c, d)	**16.** (a, d)	**17.** (a, d)	**18.** (b, c, d)
19. (a, d)	**20.** (a, b)	**21.** (a, c)	**22.** (a, c, d)

Matching Column Type

23. (a → p), (b → s), (c → q), (d → r)

24. (a → q, s), (b → s), (c → r), (d → p, r)

Numerical Value Type

25. (6) **26.** (7) **27.** (8)

Linked Comprehension Type

28. (c) **29.** (a) **30.** (b)

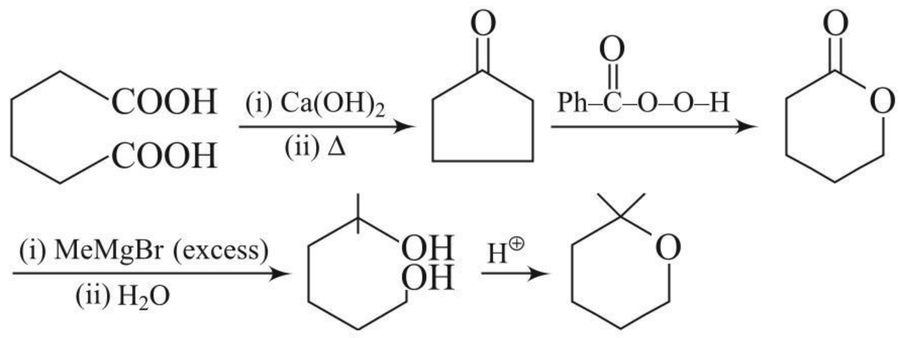
Arihant Gath

CHAPTER 1

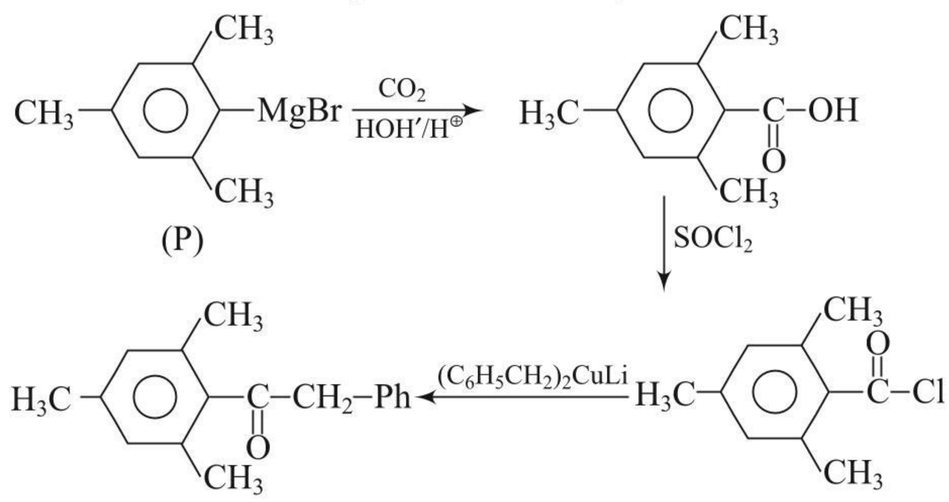
DPP 1.1

Single Correct Answer Type

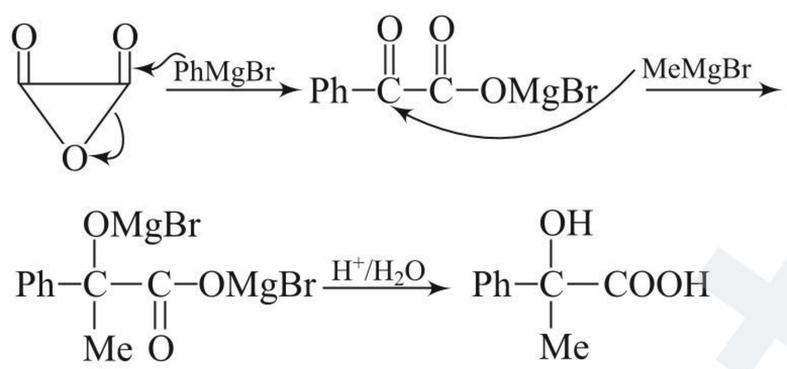
1. (a)



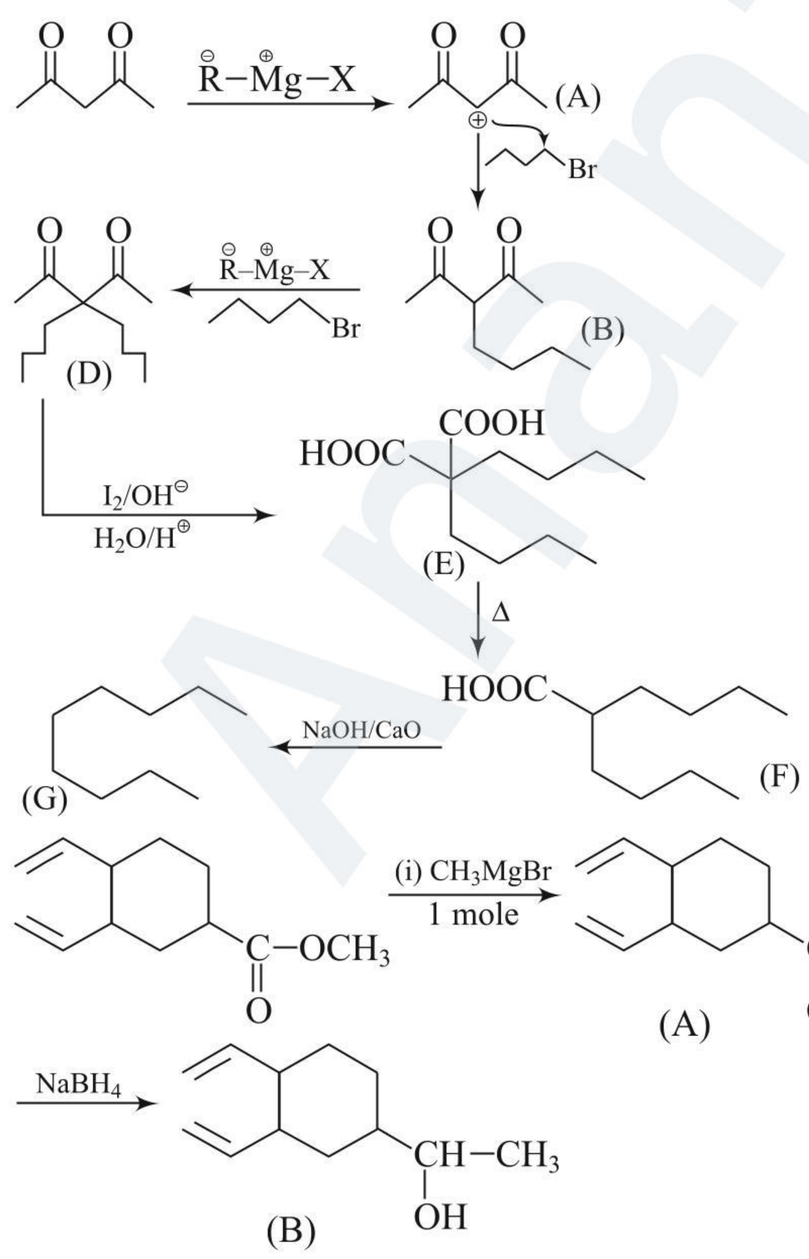
2. (d)



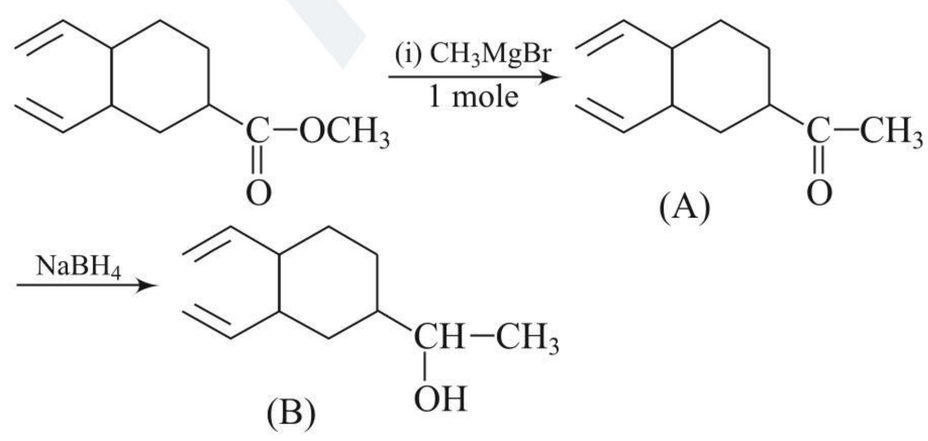
3. (c)



4. (a)

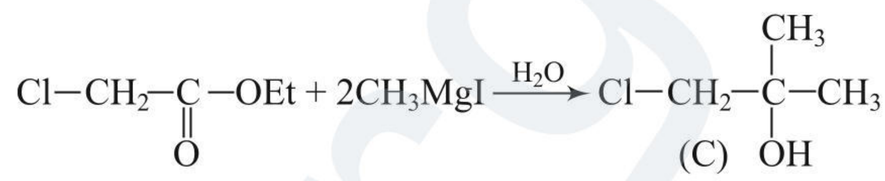


5. (a)

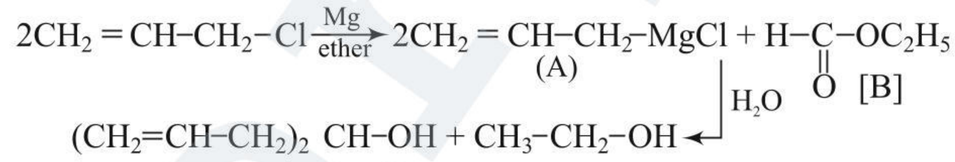


6. (c) Organomagnesium and organolithium compound cannot be prepared from the alkyl (or aryl) halide having $-NO_2$ group. On the other hand, organocopper and organocadmium compounds, do not react with the $-NO_2$ group.

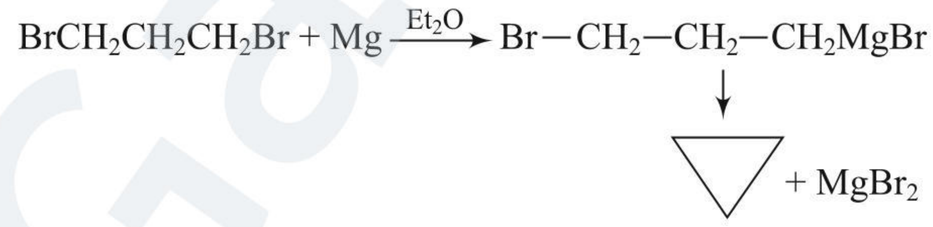
7. (c)



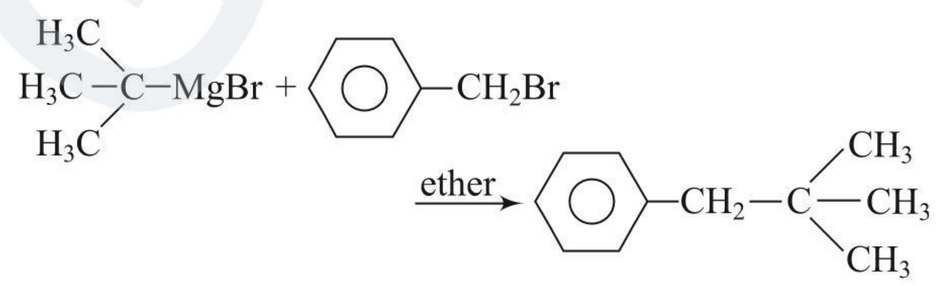
8. (b)



9. (d)



10. (b)

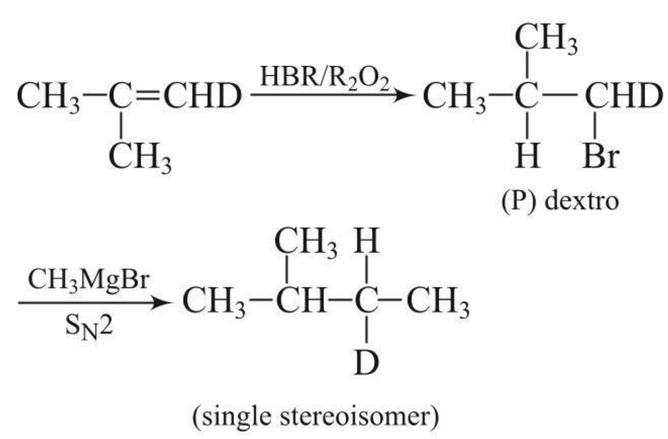


11. (d) Only methyllithium has C – M bond.

12. (c) Grignard reagents are strongly basic in character due to carbanionic nature, hence it will react with water, a weak acid.

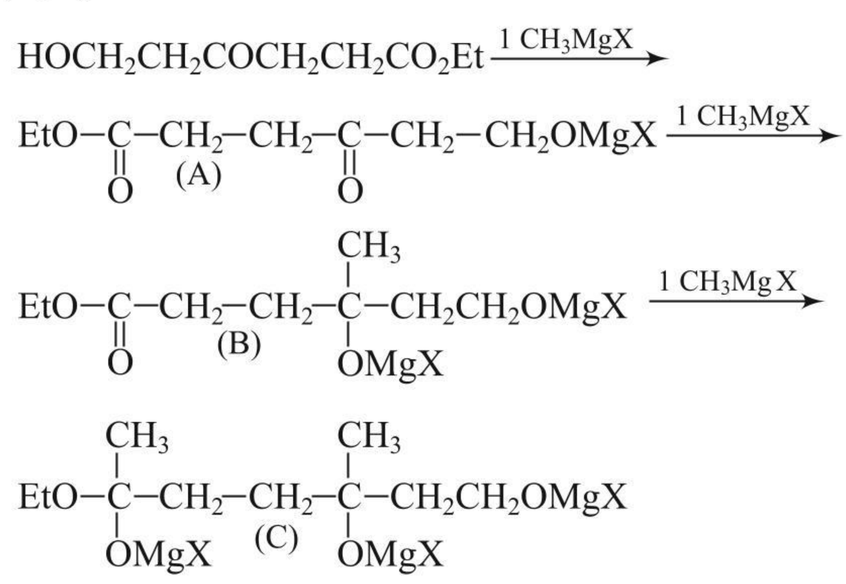
13. (b) Tertiary alcohols are formed by treating Grignard reagents either with ketones or excess of an ester other than formate which will give 2° alcohol.

14. (c)

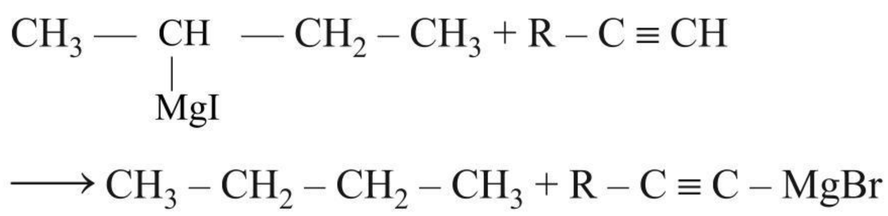


Multiple Correct Answers Type

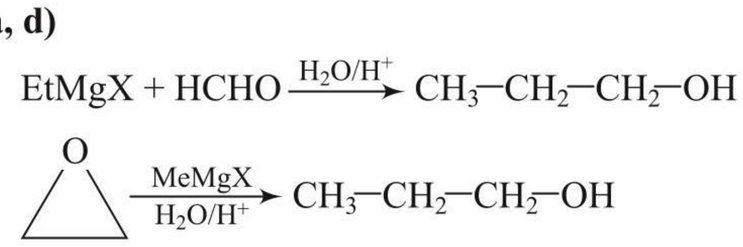
15. (a, c, d)



16. (a, d)



17. (a, d)



18. (b, c, d)

19. (a, d)

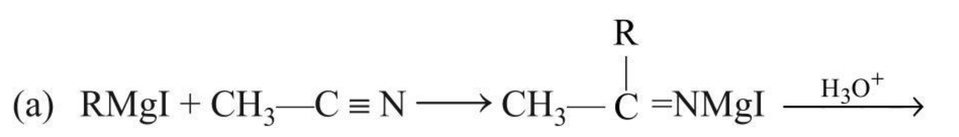
20. (a, b)

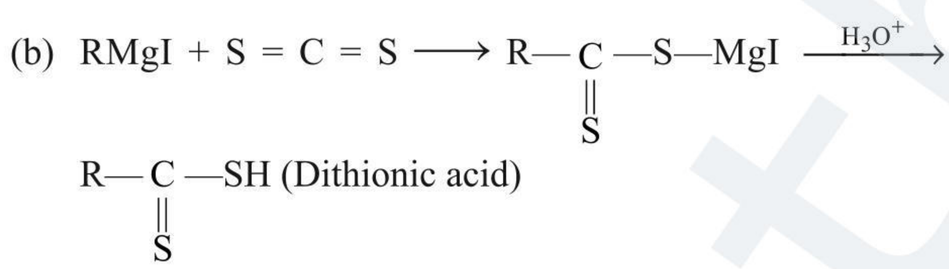
21. (a, c)

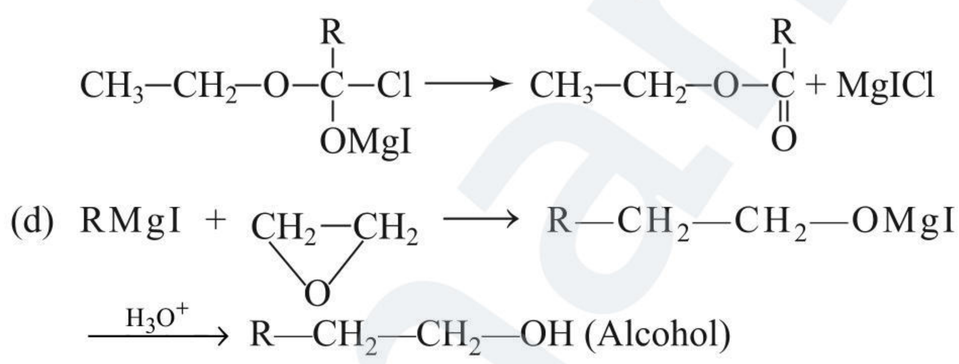
22. (a, c, d)

Matching Column Type

23. (a → p), (b → s), (c → q), (d → r)



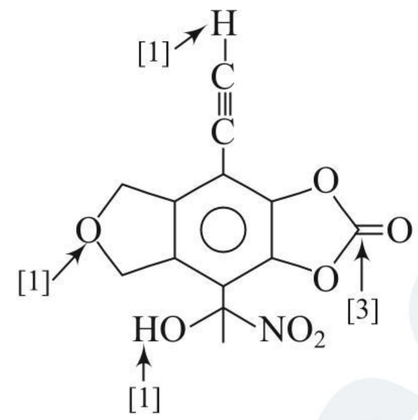




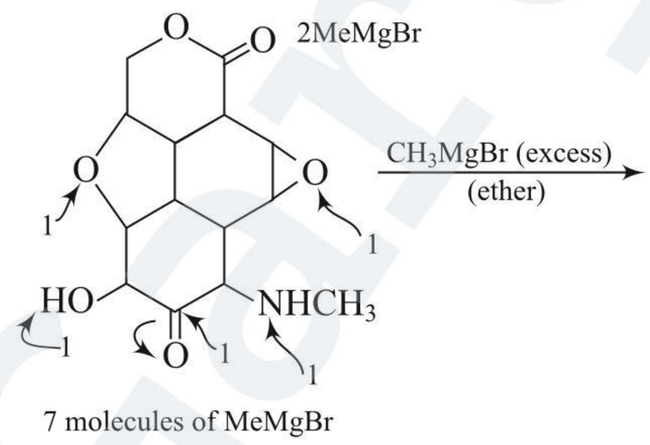
24. (a → q, s), (b → s), (c → r), (d → p, r)

Numerical Value Type

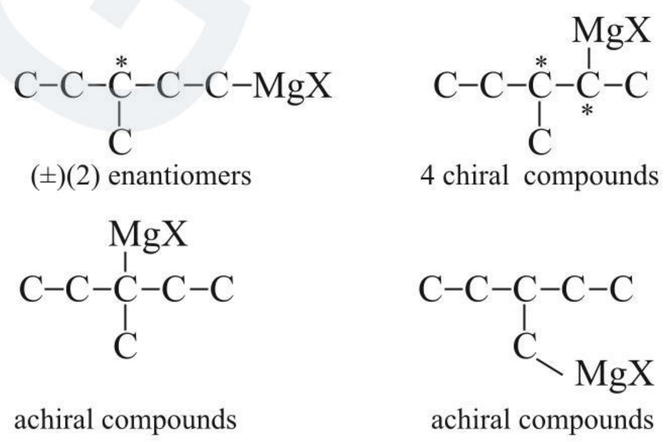
25. (6)



26. (7)

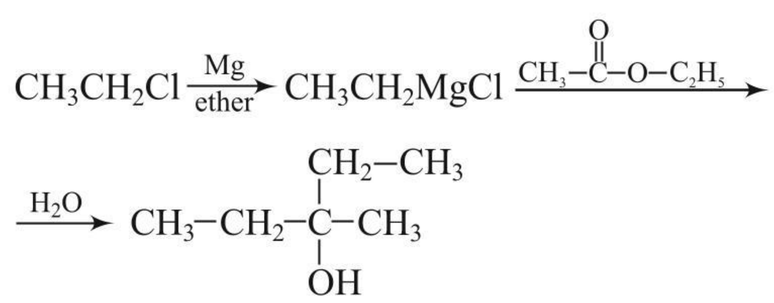


27. (8)

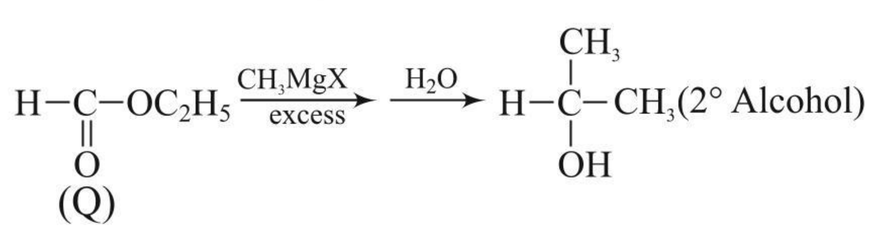
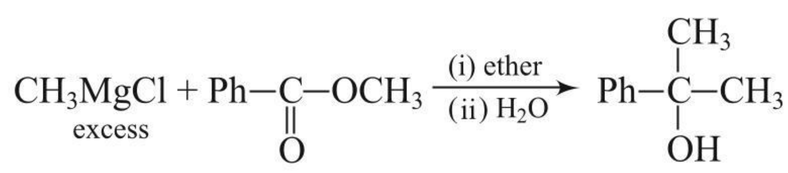


Linked Comprehension Type

28. (c)



29. (a)



30. (b)